

Retail Customer Segmentation using RFM, PCA and K-Means Clustering

A Project Report

Submitted by

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1 Introduction

Customer segmentation is a **crucial technique** in data analysis that helps businesses understand **customer behavior** and improve decision-making. By analyzing purchasing patterns, organizations can divide customers into different groups and design **targeted marketing strategies**.

In this project, customer segmentation is performed using **RFM (Recency, Frequency, Monetary)** analysis along with **K-Means clustering**. The dataset is cleaned and processed to extract meaningful features that represent customer behavior.

Further, **Principal Component Analysis (PCA)** is applied to reduce the dimensionality of the data and enable effective visualization of customer segments. This helps in clearly identifying different groups such as **high-value, loyal, regular, and inactive customers**.

The insights obtained from this analysis assist businesses in improving **customer retention**, increasing **revenue**, and making **data-driven decisions**.

2 Objective

The main objectives of this project are:

- To segment customers based on their purchasing behavior using **RFM analysis**.
- To apply **K-Means clustering** to group similar customers.
- To identify **high-value, loyal, regular, and inactive customers**.
- To visualize customer segments using **PCA and Seaborn**.
- To generate insights that support **better business decisions**.

3 Methodology

3.1 Data Collection

The dataset used in this project is obtained from the UCI Machine Learning Repository. It contains transactional data including invoice number, product details, quantity, price, customer ID, and country.

3.2 Data Cleaning

Data preprocessing was performed to ensure data quality. Missing values in **CustomerID** were removed. Transactions with negative **Quantity** and zero **UnitPrice** were filtered out, as they do not represent valid purchases.

3.3 Feature Engineering (RFM Analysis)

RFM analysis was used to transform transactional data into meaningful customer-level features:

- **Recency**: Number of days since the last purchase
- **Frequency**: Total number of transactions
- **Monetary**: Total amount spent by the customer

3.4 Data Preprocessing

Feature scaling was applied using standardization to normalize the RFM values. This step is important because K-Means clustering is sensitive to the scale of data.

3.5 K-Means Clustering

K-Means clustering algorithm was applied to segment customers into different groups based on their RFM values. Customers with similar behavior were grouped together into clusters.

3.6 Principal Component Analysis (PCA)

PCA was used to reduce the dimensionality of the dataset from three features to two principal components. This helps in visualizing the clusters effectively while retaining most of the data variance.

3.7 Visualization

Seaborn and Matplotlib were used to visualize the results. PCA scatter plots, count plots, and boxplots were used to analyze and interpret customer segments.

4 Code and Implementation Details

4.1 Loading the Dataset

```
import pandas as pd
df = pd.read_excel("OnlineRetail.xlsx")
```

The dataset is loaded using Pandas for further processing.

4.2 Data Cleaning

```
df = df.dropna(subset=["CustomerID"])
df = df[df["Quantity"] > 0]
df = df[df["UnitPrice"] > 0]
```

Invalid and missing values are removed to ensure data quality.

4.3 RFM Feature Creation

```
df["TotalAmount"] = df["Quantity"] * df["UnitPrice"]
```

RFM features are calculated to represent customer behavior.

4.4 Scaling

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
rfm_scaled = scaler.fit_transform(rfm)
```

Feature scaling is applied to normalize the data.

4.5 K-Means Clustering

```
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=4)
rfm["Cluster"] = kmeans.fit_predict(rfm_scaled)
```

Customers are grouped into clusters based on similarity.

4.6 PCA

```
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
rfm_pca = pca.fit_transform(rfm_scaled)
```

PCA is used to reduce dimensions for visualization.

5 Results and Observations

5.1 Customer Segmentation Visualization

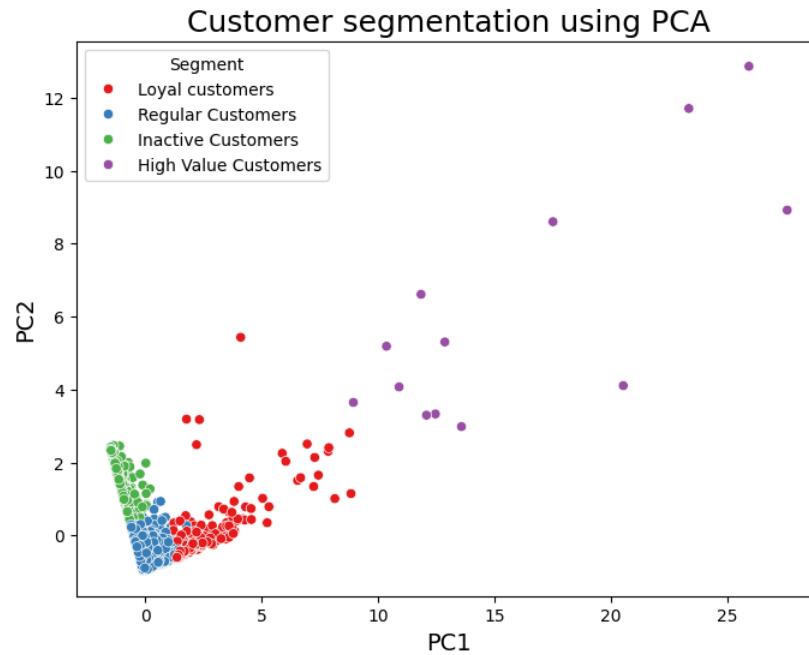


Figure 1: Customer Segmentation using PCA

The PCA scatter plot shows that customers are grouped into distinct clusters. Although slight overlap exists, the segmentation is reasonably clear, indicating that the clustering model is effective.

5.2 Customer Distribution

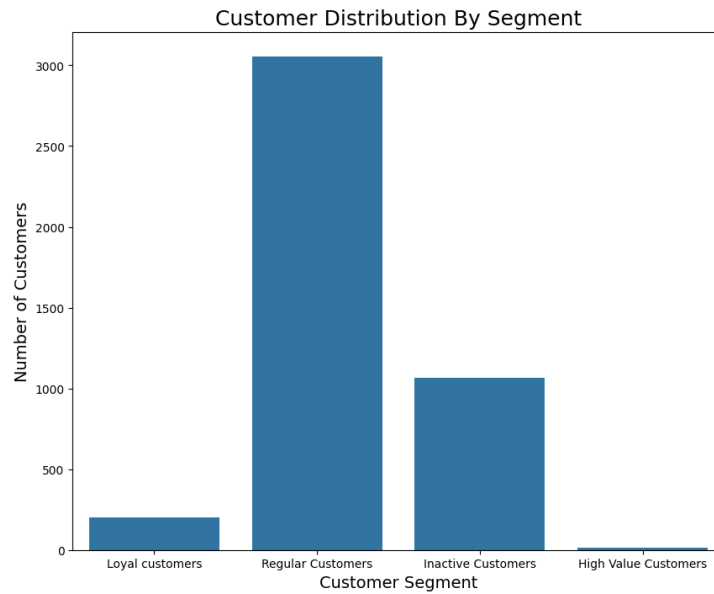


Figure 2: Customer Distribution by Segment

The count plot highlights the number of customers in each segment, helping identify which customer group is dominant and which groups are smaller but potentially more valuable.

5.3 RFM Analysis

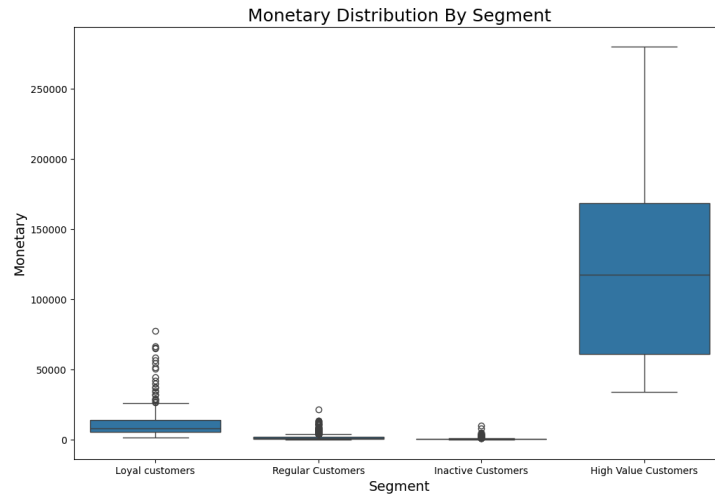


Figure 3: Monetary Distribution by Segment

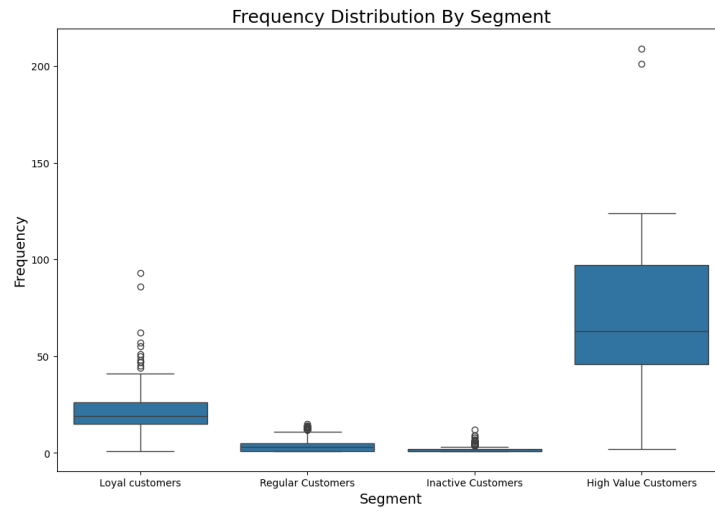


Figure 4: Frequency Distribution by Segment

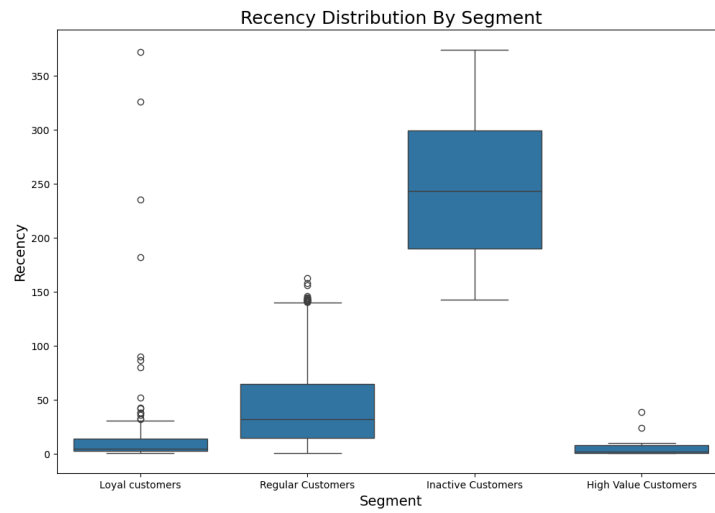
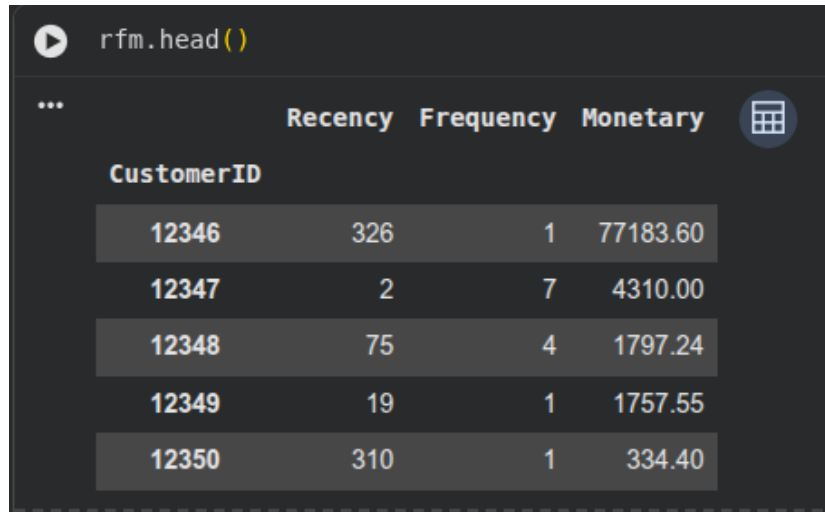


Figure 5: Recency Distribution by Segment

The boxplots clearly show variation in customer behavior across segments. Certain clusters exhibit higher spending and frequency, while others show inactivity.

5.4 RFM Table (Sample)



```
rfm.head()
```

...	Recency	Frequency	Monetary	
CustomerID				
12346	326	1	77183.60	
12347	2	7	4310.00	
12348	75	4	1797.24	
12349	19	1	1757.55	
12350	310	1	334.40	

Figure 6: Sample RFM Table

5.5 Customer Segments

Cluster	Segment
0	Regular
1	Inactive
2	High Value
3	Loyal

5.6 Observations

- **High Value customers** show high monetary value, frequent purchases, and low recency, making them the most valuable segment.

- **Loyal customers** purchase regularly but spend less compared to high-value customers.
- **Regular customers** exhibit moderate purchasing behavior.
- **Inactive customers** have high recency and low frequency, indicating low engagement.

6 Conclusion

In this project, customer segmentation was performed using RFM analysis and K-Means clustering. The dataset was cleaned and transformed into meaningful features to represent customer behavior.

The analysis successfully identified different customer segments such as high-value, loyal, regular, and inactive customers. These segments clearly show variations in purchasing patterns and engagement levels.

The insights obtained from this project can help businesses implement targeted marketing strategies, improve customer retention, and make better data-driven decisions.

7 Dataset and Files Used

Dataset Source:

UCI Machine Learning Repository - Online Retail Dataset

Link:

<https://archive.ics.uci.edu/ml/datasets/Online+Retail>

Files Used:

- OnlineRetail.xlsx
- cleaned_online_retail.csv
- Retail_Customer_Clustering.ipynb